

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type:	Audio Processor
Manufacturer:	ClearOne
Firmware Version:	8.2.12.0
Software Version:	7.3.0.16 (CONSOLE AI)
Model(s):	Converge Pro 2 128TD, Converge Pro 2 128V, Converge Pro 2 128VD, Converge Pro 2 120, Converge Pro 2 48T, Beamforming Microphone Array 2, Converge Pro 2 128D, Converge Pro 2 128SR, Converge Pro 2 128SRD, Converge Pro 2 48V, Converge Pro 2 48VT, Converge Pro 2 128VT

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.14.0000-b007	2.13.0

Version History

Module Version	Date	Notes
1_14_0_0	10/6/2021	- Updated Matrix Tie Command: Added 'Crosspoint Type' and 'Value' parameters. - Added commands: Beam Report, Gate Report, and Room Select. - Updated Beamforming Microphone Array Zone command for Beamforming Microphone Array 2 model: Added support for zones 7 to 12.
1_13_1_0	1/15/2020	- Added query delay logic to block any queries being sent to the device after specific commands are executed. (See module notes for more details) - Split the commands for VoIP and Telco models: - DTMF → DTMFTelco, DTMFVoIP - Hook → HookTelco, HookVoIP

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		• Renamed command ○ InitiateOrAddtoConferenceCall →InitiateorAddtoConferenceCall • Updated CallDirection status. • Added 'Resume' value to HoldTransfer command. • Fixed CallState status Removed support for Party Line '0' for HookVoIP and CallState commands.
1_10_4_0	12/13/2018	• Added model: Converge Pro 2 48V Updated module based on the latest firmware 5.0.38.0
1_0_1_0	1/5/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword and deviceUsername must be set accordingly. devicePassword and deviceUsername default values are converge and clearone respectively.
Example: `InterfaceName.devicePassword = 'extron'`
`InterfaceName.devicePassword = 'converge'`
- NumberofSkypeSearch variable must be set accordingly. Default value is '5'. NumberofSkypeSearch ranges from 1 to 10.
Example: `InterfaceName.NumberofSkypeSearch = '5'`
- All Skype for Business commands and functionality is only supported when using the ClearOne Pro Console v6.5.5.0 and CONVERGE Pro 2 firmware v6.0.33.0.
- This module contains built-in query delay which blocks any query from sending out to the device after specific actions.
 - Sending Username will trigger a 2 seconds query delay
 - Sending Password will trigger a 2 seconds query delay
 - Sending SkypeNavigation will trigger a 3 seconds query delay
 - Sending SkypeUpdate will trigger a 17 seconds query delay
- Do **NOT** send any queries when Firmware status is 'Initializing'.
 - Upon receiving an 'Initializing' status from the device, this module will trigger a 165 second query delay.

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='Converge Pro 2 128TD')
```

SerialOverEthernetClass

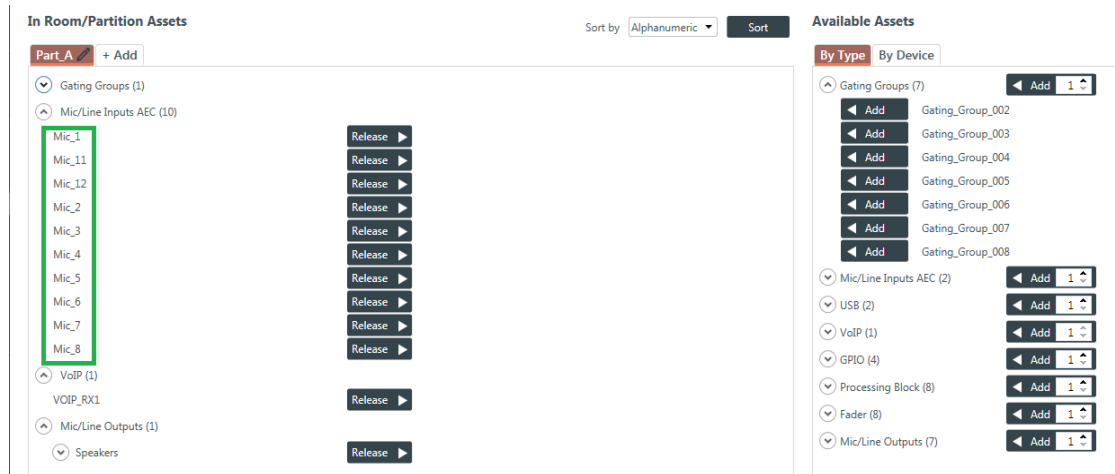
```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='Converge Pro 2 128TD')
```

EthernetClass

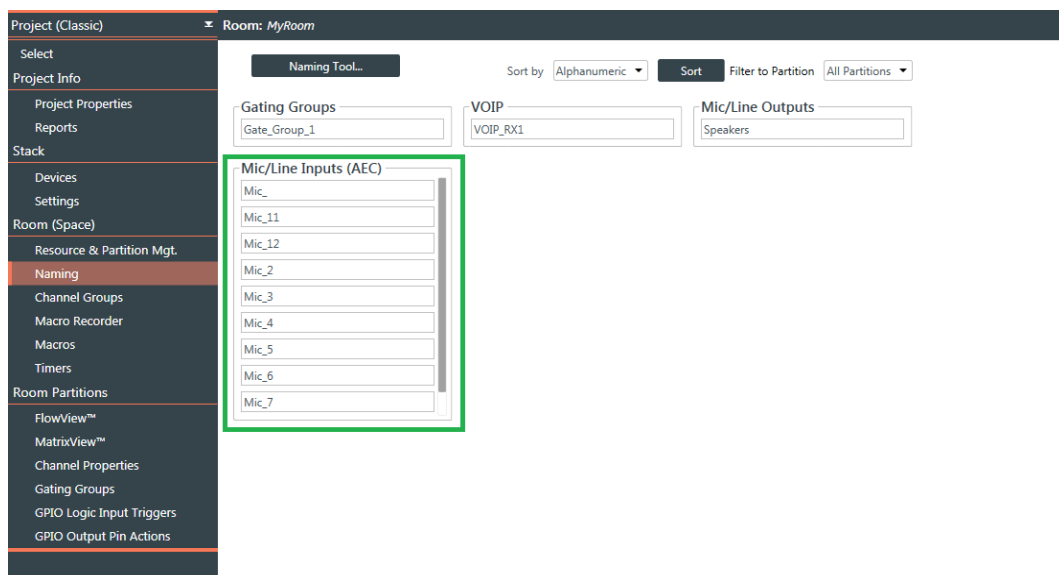
```
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='Converge Pro 2 128TD')
```

Setting the Channel Names

- Most of the commands in the module require Channel qualifier. Channel Name qualifiers must match the names that are set on the Clearone Converge Console software.
- In the example provided below, the Channel Names for Mics are named Mic_1 to Mic_12. The Clearone Converge Console software does not allow white space characters to be used when naming channel names.



- The Channel Name can be set in the Naming section of the Converge console software.



Setting up a Conference Call

- Dial the first number which will be Party Line 1.
- Select Party Line 2 using the Select Party Line command. This will place the Party Line 1 on Hold.
- Dial the second number.
- Using the Initiate or Add to Conference Call command you can take the Party Line 1 off hold.
- Use the Join Conference command to bring both calls together in a conference.
- Use the On-Hook command to end the conference call.

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command BeamformingMicrophone ArrayGain ¹	Value -65 to 20 in steps of 0.5 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# BeamformingMicrophoneArrayGain example InterfaceName.Set('BeamformingMicrophoneArrayGain', 20, {'Channel Name': 'String'})			
Command BeamformingMicrophone ArrayMode ¹	Value 'Auto'	Value 'Ceiling'	Value 'Wall'
	'Tabletop'		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# BeamformingMicrophoneArrayMode example InterfaceName.Set('BeamformingMicrophoneArrayMode', 'Auto', {'Channel Name': 'String'})			
Command BeamformingMicrophone ArrayMute ¹	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# BeamformingMicrophoneArrayMute example InterfaceName.Set('BeamformingMicrophoneArrayMute', 'On', {'Channel Name': 'String'})			
Command BeamformingMicrophone ArrayMuteLED ¹	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# BeamformingMicrophoneArrayMuteLED example InterfaceName.Set('BeamformingMicrophoneArrayMuteLED', 'On', {'Channel Name': 'String'})			
Command BeamformingMicrophone ArrayZone ¹	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Zone'	Qualifier Value '1' – '12'		
# BeamformingMicrophoneArrayZone example InterfaceName.Set('BeamformingMicrophoneArrayZone', 'On', {'Channel Name': 'String', 'Zone': '1'})			

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Command CallForwarding ^{2, 4}	Value 'Disable' 'No Answer'	Value 'Unconditional'	Value 'Busy'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# CallForwarding example InterfaceName.Set('CallForwarding', 'Disable', {'Channel Name': 'String'})			
Command ClearMatrixTies ²	Value None		
# ClearMatrixTies example InterfaceName.Set('ClearMatrixTies', None)			
Command DoNotDisturb ^{2, 4}	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# DoNotDisturb example InterfaceName.Set('DoNotDisturb', 'On', {'Channel Name': 'String'})			
Command DTMFTelco ^{2, 5}	Value '0' – '9'	Value '*'	Value '#'
Qualifier Key 'Box Number'	Qualifier Value '1' – '9'		
# DTMFTelco example InterfaceName.Set('DTMFTelco', '0', {'Box Number': '1'})			
Command DTMFVoIP ^{2, 4}	Value '0' – '9'	Value '*'	Value '#'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# DTMFVoIP example InterfaceName.Set('DTMFVoIP', '0', {'Channel Name': 'String'})			
Command Gain ²	Value -65 to 20 in steps of 0.5 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# Gain example InterfaceName.Set('Gain', 20, {'Channel Name': 'String'})			
Command GainCoarse ²	Value 0 to 56 in steps of 7 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# GainCoarse example InterfaceName.Set('GainCoarse', 56, {'Channel Name': 'String'})			
Command GainFine ²	Value -65 to 20 in steps of 0.5 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# GainFine example InterfaceName.Set('GainFine', 20, {'Channel Name': 'String'})			
Command GraphicEQEnable ²	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# GraphicEQEnable example InterfaceName.Set('GraphicEQEnable', 'On', {'Channel Name': 'String'})			

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Command GraphicEQGain ²	Value -12 to 12 in steps of 1 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Gain Number'	Qualifier Value '1' – '10'		
# GraphicEQGain example InterfaceName.Set('GraphicEQGain', 12, {'Channel Name': 'String', 'Gain Number': '1'})			
Command HoldTransfer ^{2, 4}	Value 'Hold' 'Blind Transfer'	Value 'Resume'	Value 'Transfer'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# HoldTransfer example InterfaceName.Set('HoldTransfer', 'Hold', {'Channel Name': 'String'})			
Command HookTelco ^{2, 5}	Value 'On Hook' 'Redial'	Value 'Off Hook' 'Flash'	Value 'Dial'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Dial String' ⁹	Qualifier Value 'String'		
# HookTelco example – On Hook, Off Hook, Redial, and Flash InterfaceName.Set('HookTelco', 'On Hook', {'Channel Name': 'String'}) # HookTelco example – Dial InterfaceName.Set('HookTelco', 'Dial', {'Channel Name': 'String', 'Dial String': 'String'})			
Command HookVoIP ^{2, 4}	Value 'On Hook' 'Redial'	Value 'Off Hook'	Value 'Dial'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Dial String' ⁹	Qualifier Value 'String'		
# HookVoIP example – On Hook, Off Hook, Redial, and Flash InterfaceName.Set('HookVoIP', 'On Hook', {'Channel Name': 'String'}) # HookVoIP example – Dial InterfaceName.Set('HookVoIP', 'On Hook', {'Channel Name': 'String', 'Dial String': 'String'})			
Command InitiateorAddtoConferenc eCall ^{2, 4}	Value '0' – '5'		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# InitiateorAddtoConferenceCall example InterfaceName.Set('InitiateorAddtoConferenceCall', '0', {'Channel Name': 'String'})			
Command JoinConference ^{2, 4}	Value None		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# JoinConference example InterfaceName.Set('JoinConference', None, {'Channel Name': 'String'})			

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Command Macro ²	Value None		
Qualifier Key 'Name'	Qualifier Value 'String'		
# Macro example InterfaceName.Set('Macro', {'Name': 'String'})			
Command MatrixTie ^{2, 8}	Value 'Tie'	Value 'Untie'	
Qualifier Key 'Input'	Qualifier Value 'String'		
Qualifier Key 'Output'	Qualifier Value 'String'		
Qualifier Key 'Crosspoint Type'	Qualifier Value 'Normal' 'Pre-AEC' ⁷	Qualifier Value 'Gated' ⁷	Qualifier Value 'Non-Gated' ⁷
# MatrixTie example InterfaceName.Set('MatrixTie', 'Tie', {'Input': 'String', 'Output': 'String', 'Crosspoint Type': 'Normal'})			
Command MicPhantomPower ²	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# MicPhantomPower example InterfaceName.Set('MicPhantomPower', 'On', {'Channel Name': 'String'})			
Command Mute ²	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# Mute example InterfaceName.Set('Mute', 'On', {'Channel Name': 'String'})			
Command OutputMicLine ²	Value 'Line Level'	Value 'Mic Level'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# OutputMicLine example InterfaceName.Set('OutputMicLine', 'Line Level', {'Channel Name': 'String'})			
Command OutputPolarity ²	Value 'Default'	Value 'Reverse'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# OutputPolarity example InterfaceName.Set('OutputPolarity', 'Default', {'Channel Name': 'String'})			
Command PostGain ²	Value 0 to 20 in steps of 0.5 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# PostGain example InterfaceName.Set('PostGain', 20, {'Channel Name': 'String'})			

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Command Reject ^{2,4}	Value 'Line 1' 'Line 4'	Value 'Line 2' 'Line 5'	Value 'Line 3'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# Reject example InterfaceName.Set('Reject', 'Line 1', {'Channel Name': 'String'})			
Command RoomSelect ²	Value None		
Qualifier Key 'Room Number'	Qualifier Value Number		
Qualifier Key 'Sub Room'	Qualifier Value 'String'		
Qualifier Key 'Config File'	Qualifier Value 'String'		
# RoomSelect example InterfaceName.Set('RoomSelect', None, {'Room Number': , 'Sub Room': 'String', 'Config File': 'String'})			
Command SelectPartyLine ^{2,4}	Value '1' – '5'		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# SelectPartyLine example InterfaceName.Set('SelectPartyLine', '1', {'Channel Name': 'String'})			
Command SignalGeneratorEnable ²	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# SignalGeneratorEnable example InterfaceName.Set('SignalGeneratorEnable', 'On', {'Channel Name': 'String'})			
Command SkypeNavigation ^{2,4}	Value 'Up' 'Page Down'	Value 'Down'	Value 'Page Up'
# SkypeNavigation example InterfaceName.Set('SkypeNavigation', 'Up')			
Command SkypeSearch ^{2,4}	Value 'String'		
# SkypeSearch example InterfaceName.Set('SkypeSearch', 'String')			
Command SkypeSearchSet ^{2,4}	Value 1 – 10		
# SkypeSearchSet example InterfaceName.Set('SkypeSearchSet', 10)			
Command SkypeUpdate ^{2,4}	Value None		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Search Name'	Qualifier Value 'String'		
# SkypeUpdate example InterfaceName.Set('SkypeUpdate', None, {'Channel Name': 'String', 'Search Name': 'String'})			

¹ Command is only available on Beamforming Microphone Array 2.

² Command is not available on Beamforming Microphone Array 2.

³ Skype Update reads the string entered on Skype Search to populate status on Skype Search Result.

⁴ Only supported for VoIP models.

⁵ Only supported for Telco models.

⁶ This parameter is not required when using the command.

⁷ Only supported for MIC end points.

⁸ You shouldn't link a channel to itself, or to its input/output equivalent, or otherwise create similar loops. For example, you should not connect TELCO_RX 1 to TELCO_TX 1.

⁹ Dial String qualifier is only needed for 'Dial' command

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus, IncomingCallStatus, and SkypeSearchResult do not support the Update function. ConnectionStatus is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
```

FeedbackHandler will be called when any qualifier gets a new status.

Command BeamReport ⁸	Value 'Active'	Value 'Not Active'
Qualifier Key 'Device Name' ⁶	Qualifier Value 'String'	
Qualifier Key 'BMA360 ID' ⁷	Qualifier Value 'String'	
Qualifier Key 'Beam'	Qualifier Value '1' – '12'	
<pre># BeamReport example InterfaceName.Update('BeamReport', {'Device Name': 'String', 'BMA360 ID': 'String', 'Beam': '1'}) Value = InterfaceName.ReadStatus('BeamReport', {'Device Name': 'String', 'BMA360 ID': 'String', 'Beam': '1'}) InterfaceName.SubscribeStatus('BeamReport', None, FeedbackHandler)</pre>		
Command BeamformingMicrophone ArrayGain ¹	Value -65 to 20 in steps of 0.5 dB	
Qualifier Key 'Channel Name'	Qualifier Value 'String'	
<pre># BeamformingMicrophoneArrayGain example InterfaceName.Update('BeamformingMicrophoneArrayGain', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('BeamformingMicrophoneArrayGain', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('BeamformingMicrophoneArrayGain', None, FeedbackHandler)</pre>		
Command BeamformingMicrophone ArrayMode ¹	Value 'Auto'	Value 'Ceiling'
	Value 'Wall'	
	Value 'Tabletop'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'	
<pre># BeamformingMicrophoneArrayMode example InterfaceName.Update('BeamformingMicrophoneArrayMode', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('BeamformingMicrophoneArrayMode', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('BeamformingMicrophoneArrayMode', None, FeedbackHandler)</pre>		

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Command BeamformingMicrophone ArrayMute ¹	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# BeamformingMicrophoneArrayMute example InterfaceName.Update('BeamformingMicrophoneArrayMute', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('BeamformingMicrophoneArrayMute', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('BeamformingMicrophoneArrayMute', None, FeedbackHandler)			
Command BeamformingMicrophone ArrayMuteLED ¹	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# BeamformingMicrophoneArrayMuteLED example InterfaceName.Update('BeamformingMicrophoneArrayMuteLED', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('BeamformingMicrophoneArrayMuteLED', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('BeamformingMicrophoneArrayMuteLED', None, FeedbackHandler)			
Command BeamformingMicrophone ArrayZone ¹	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Zone'	Qualifier Value '1' – '12'		
# BeamformingMicrophoneArrayZone example InterfaceName.Update('BeamformingMicrophoneArrayZone', {'Channel Name': 'String', 'Zone': '1'}) Value = InterfaceName.ReadStatus('BeamformingMicrophoneArrayZone', {'Channel Name': 'String', 'Zone': '1'}) InterfaceName.SubscribeStatus('BeamformingMicrophoneArrayZone', None, FeedbackHandler)			
Command CallDirection ^{2, 4}	Value 'Incoming'	Value 'Outgoing'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# CallDirection example InterfaceName.Update('CallDirection', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('CallDirection', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('CallDirection', None, FeedbackHandler)			
Command CallForwarding ^{2, 3}	Value 'Disable' 'No Answer'	Value 'Unconditional'	Value 'Busy'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# CallForwarding example InterfaceName.Update('CallForwarding', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('CallForwarding', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('CallForwarding', None, FeedbackHandler)			
Command CallState ^{2, 3}	Value 'Idle' 'In Process' 'Active' 'In Conference Call,	Value 'Dial Tone' 'Ringing' 'Hold' 'On Hold In Conference	Value 'Dialing' 'Busy' 'Incoming' 'Transferring, Call On

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	Active' 'Transferring, Active Call' 'Transferring, In Process' 'Transferring, Active' 'Blind Transfer, Dial Tone' 'Blind Transfer, Ringing'	Call' 'Transferring, Dial Tone' 'Transferring, Ringing' 'Transferring, On Hold' 'Blind Transfer, Dialing' 'Blind Transfer, Busy'	Hold' 'Transferring, Dialing' 'Transferring, Busy' 'Blind Transfer, On Hold' 'Blind Transfer, In Process'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Party Line'	Qualifier Value '1' – '5'		
# CallState example InterfaceName.Update('CallState', {'Channel Name': 'String', 'Party Line': '1'}) Value = InterfaceName.ReadStatus('CallState', {'Channel Name': 'String', 'Party Line': '1'}) InterfaceName.SubscribeStatus('CallState', None, FeedbackHandler)			
Command CallerID ^{2,3}	Value 'String'		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# CallerID example InterfaceName.Update('CallerID', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('CallerID', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('CallerID', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DoNotDisturb ^{2,3}	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# DoNotDisturb example InterfaceName.Update('DoNotDisturb', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('DoNotDisturb', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('DoNotDisturb', None, FeedbackHandler)			
Command Firmware	Value 'String'		
# Firmware example InterfaceName.Update('Firmware') Value = InterfaceName.ReadStatus('Firmware') InterfaceName.SubscribeStatus('Firmware', None, FeedbackHandler)			
Command Gain ²	Value -65 to 20 in steps of 0.5 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# Gain example InterfaceName.Update('Gain', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('Gain', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('Gain', None, FeedbackHandler)			

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Command GainCoarse ²	Value 0 to 56 in steps of 7 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# GainCoarse example InterfaceName.Update('GainCoarse', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('GainCoarse', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('GainCoarse', None, FeedbackHandler)			
Command GainFine ²	Value -65 to 20 in steps of 0.5 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# GainFine example InterfaceName.Update('GainFine', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('GainFine', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('GainFine', None, FeedbackHandler)			
Command GateReport ²	Value 'Gated'	Value 'Not Gated'	
Qualifier Key 'Channel'	Qualifier Value '1' – '4'	Qualifier Value '5' – '12' ⁵	
Qualifier Key 'Device Name'	Qualifier Value 'String'		
# GateReport example InterfaceName.Update('GateReport', {'Channel': '1', 'Device Name': 'String'}) Value = InterfaceName.ReadStatus('GateReport', {'Channel': '1', 'Device Name': 'String'}) InterfaceName.SubscribeStatus('GateReport', None, FeedbackHandler)			
Command GraphicEQEnable ²	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# GraphicEQEnable example InterfaceName.Update('GraphicEQEnable', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('GraphicEQEnable', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('GraphicEQEnable', None, FeedbackHandler)			
Command GraphicEQGain ²	Value -12 to 12 in steps of 1 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Gain Number'	Qualifier Value '1' – '10'		
# GraphicEQGain example InterfaceName.Update('GraphicEQGain', {'Channel Name': 'String', 'Gain Number': '1'}) Value = InterfaceName.ReadStatus('GraphicEQGain', {'Channel Name': 'String', 'Gain Number': '1'}) InterfaceName.SubscribeStatus('GraphicEQGain', None, FeedbackHandler)			
Command HookTelco ^{2,4}	Value 'On Hook' 'Redial'	Value 'Off Hook' 'Flash'	Value 'Dial'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# HookTelco example InterfaceName.Update('HookTelco', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('HookTelco', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('HookTelco', None, FeedbackHandler)			

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Command HookVoIP ^{2,3}	Value 'On Hook' 'Redial'	Value 'Off Hook'	Value 'Dial'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Party Line'	Qualifier Value '1' – '5'		
# HookVoIP example InterfaceName.Update('HookVoIP', {'Channel Name': 'String', 'Party Line': '1'}) Value = InterfaceName.ReadStatus('HookVoIP', {'Channel Name': 'String', 'Party Line': '1'}) InterfaceName.SubscribeStatus('HookVoIP', None, FeedbackHandler)			
Command IncomingCallStatus ^{2,4}	Value 'No Call'	Value 'Call Incoming'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# IncomingCallStatus example Value = InterfaceName.ReadStatus('IncomingCallStatus', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('IncomingCallStatus', None, FeedbackHandler)			
Command Mute ²	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# Mute example InterfaceName.Update('Mute', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('Mute', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('Mute', None, FeedbackHandler)			
Command OutputMicLine ²	Value 'Line Level'	Value 'Mic Level'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# OutputMicLine example InterfaceName.Update('OutputMicLine', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('OutputMicLine', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('OutputMicLine', None, FeedbackHandler)			
Command OutputPolarity ²	Value 'Default'	Value 'Reverse'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# OutputPolarity example InterfaceName.Update('OutputPolarity', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('OutputPolarity', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('OutputPolarity', None, FeedbackHandler)			
Command PostGain ²	Value 0 to 20 in steps of 0.5 dB		
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# PostGain example InterfaceName.Update('PostGain', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('PostGain', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('PostGain', None, FeedbackHandler)			

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Command SignalGeneratorEnable ²	Value 'On'	Value 'Off'	
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
# SignalGeneratorEnable example InterfaceName.Update('SignalGeneratorEnable', {'Channel Name': 'String'}) Value = InterfaceName.ReadStatus('SignalGeneratorEnable', {'Channel Name': 'String'}) InterfaceName.SubscribeStatus('SignalGeneratorEnable', None, FeedbackHandler)			
Command SkypeCallStatus ^{2,3}	Value 'Idle' 'Active' 'Join Conference' 'No Active Session'	Value 'Ringing' 'Hold' 'Add Audio'	Value 'Busy' 'Incoming' 'Connecting'
Qualifier Key 'Channel Name'	Qualifier Value 'String'		
Qualifier Key 'Session ID'	Qualifier Value '0' – '5'		
# SkypeCallStatus example InterfaceName.Update('SkypeCallStatus', {'Channel Name': 'String', 'Session ID': '0'}) Value = InterfaceName.ReadStatus('SkypeCallStatus', {'Channel Name': 'String', 'Session ID': '0'}) InterfaceName.SubscribeStatus('SkypeCallStatus', None, FeedbackHandler)			
Command SkypeSearchResult ^{2,3}	Value 'String'		
Qualifier Key 'Button'	Qualifier Value 1 to 10 in steps of 1		
# SkypeSearchResult example Value = InterfaceName.ReadStatus('SkypeSearchResult', {'Button': 10}) InterfaceName.SubscribeStatus('SkypeSearchResult', None, FeedbackHandler)			

¹ Command is only available on Beamforming Microphone Array 2

² Command is not available on Beamforming Microphone Array 2

³ Only supported for VoIP models.

⁴ Only supported for Telco models.

⁵ Only supported on 12 channel models.

⁶ 'Device Name' parameter is obtained from the device.

⁷ This value is based on how many Daisy Chained BMA360 devices are used. First device starts with '101', second device is '102', etc.

⁸ This command is only supported if a BMA360 device is connected to the ClearOne unit.

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

- Devices have different numbers of inputs, outputs, faders, microphones, and processing. Please refer to owner manual or the table in the Driver Notes section.
- If using this driver with any Extron Electronics product with only 3-wire serial control, the ClearOne device flow control must be disabled. The ClearOne device settings can be configured using the manufacturer's control software.

Serial communication

Port Type: RS-232

Baud Rate: 57600

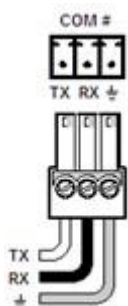
Data Bits: 8

Parity: None

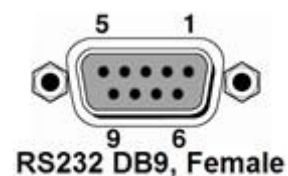
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet (TCP)

Default Port: 23

Logon Credentials Supported: Yes

Default Username: clearone

Default Password: converge

Multi-Connection Yes

Capabilities:

Port Changeability: Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication. Please refer to user manual for settings and changes to the network communication parameters such as: Username, Password and Port Number.

Notes for the Device

Appendix A. Set Commands

Beamforming Microphone Array Gain 20 Channel Name testString	EP testString LEVEL GAIN 20\x0D
Beamforming Microphone Array Gain -65 Channel Name testString	EP testString LEVEL GAIN -65\x0D
Beamforming Microphone Array Mode Auto Channel Name testString	EP testString BF BF_MODE 1\x0D
Beamforming Microphone Array Mode Ceiling Channel Name testString	EP testString BF BF_MODE 2\x0D
Beamforming Microphone Array Mode Tabletop Channel Name testString	EP testString BF BF_MODE 4\x0D
Beamforming Microphone Array Mode Wall Channel Name testString	EP testString BF BF_MODE 3\x0D
Beamforming Microphone Array Mute LED Off Channel Name testString	EP testString BF BF_LED 0\x0D
Beamforming Microphone Array Mute LED On Channel Name testString	EP testString BF BF_LED 1\x0D
Beamforming Microphone Array Mute Off Channel Name testString	EP testString LEVEL MUTE 0\x0D
Beamforming Microphone Array Mute On Channel Name testString	EP testString LEVEL MUTE 1\x0D
Beamforming Microphone Array Zone Off Channel Name testString Zone 1	EP testString BF ZONE_1 0\x0D
Beamforming Microphone Array Zone Off Channel Name testString Zone 12	EP testString BF ZONE_12 0\x0D
Beamforming Microphone Array Zone On Channel Name testString Zone 1	EP testString BF ZONE_1 1\x0D
Beamforming Microphone Array Zone On Channel Name testString Zone 12	EP testString BF ZONE_12 1\x0D
Call Forwarding Busy Channel Name testString	EP testString KEY KEY_FORWARD 2\x0D
Call Forwarding Disable Channel Name testString	EP testString KEY KEY_FORWARD 0\x0D
Call Forwarding No Answer Channel Name testString	EP testString KEY KEY_FORWARD 3\x0D
Call Forwarding Unconditional Channel Name testString	EP testString KEY KEY_FORWARD 1\x0D
Clear Matrix Ties None	CLRMATRIX\x0D
Do Not Disturb Off Channel Name testString	EP testString KEY KEY_DO_NOT_DISTURB 0\x0D
Do Not Disturb On Channel Name testString	EP testString KEY KEY_DO_NOT_DISTURB 1\x0D
DTMF VoIP # Channel Name testString	EP testString KEY KEY_DIGIT_PRESSED #\x0DEP testString KEY KEY_DIGIT_RELEASED #\x0D
DTMF VoIP * Channel Name testString	EP testString KEY KEY_DIGIT_PRESSED *\x0DEP testString KEY KEY_DIGIT_RELEASED *\x0D
DTMF VoIP 0 Channel Name testString	EP testString KEY KEY_DIGIT_PRESSED 0\x0DEP testString KEY KEY_DIGIT_RELEASED 0\x0D
DTMF VoIP 9 Channel Name testString	EP testString KEY KEY_DIGIT_PRESSED 9\x0DEP testString KEY KEY_DIGIT_RELEASED 9\x0D
Gain 20 Channel Name testString	EP testString LEVEL GAIN 20\x0D
Gain -65 Channel Name testString	EP testString LEVEL GAIN -65\x0D
Gain Coarse 0 Channel Name testString	EP testString LEVEL GAIN_COARSE 0\x0D

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Gain Coarse 56 Channel Name testString	EP testString LEVEL GAIN_COARSE 56\x0D
Gain Fine 20 Channel Name testString	EP testString LEVEL GAIN_FINE 20\x0D
Gain Fine -65 Channel Name testString	EP testString LEVEL GAIN_FINE -65\x0D
Graphic EQ Enable Off Channel Name testString	EP testString GRAPHICEQ ENABLE 0\x0D
Graphic EQ Enable On Channel Name testString	EP testString GRAPHICEQ ENABLE 1\x0D
Graphic EQ Gain 12 Channel Name testString Gain Number 1	EP testString GRAPHICEQ GAIN_1 12\x0D
Graphic EQ Gain -12 Channel Name testString Gain Number 1	EP testString GRAPHICEQ GAIN_1 -12\x0D
Graphic EQ Gain 12 Channel Name testString Gain Number 10	EP testString GRAPHICEQ GAIN_10 12\x0D
Graphic EQ Gain -12 Channel Name testString Gain Number 10	EP testString GRAPHICEQ GAIN_10 -12\x0D
Hold/Transfer Blind Transfer Channel Name testString	EP testString KEY KEY_BLIND_TRANSFER\x0D
Hold/Transfer Hold Channel Name testString	EP testString KEY KEY_HOLD\x0D
Hold/Transfer Resume Channel Name testString	EP testString KEY KEY_RESUME\x0D
Hold/Transfer Transfer Channel Name testString	EP testString KEY KEY_TRANSFER\x0D
Hook Telco Dial Channel Name testString	EP testString KEY KEY_CALL DialString\x0D
Hook Telco Flash Channel Name testString	EP testString KEY KEY_HOOK_FLASH\x0D
Hook Telco Off Hook Channel Name testString	EP testString KEY KEY_HOOK 1\x0D
Hook Telco On Hook Channel Name testString	EP testString KEY KEY_HOOK 0\x0D
Hook Telco Redial Channel Name testString	EP testString KEY KEY_REDIAL\x0D
Hook VoIP Dial Channel Name testString Party Line 1	EP testString KEY KEY_CALL DialString\x0D
Hook VoIP Dial Channel Name testString Party Line 5	EP testString KEY KEY_CALL DialString\x0D
Hook VoIP Off Hook Channel Name testString Party Line 1	EP testString KEY KEY_HOOK 1\x0D
Hook VoIP Off Hook Channel Name testString Party Line 5	EP testString KEY KEY_HOOK 1\x0D
Hook VoIP On Hook Channel Name testString Party Line 1	EP testString KEY KEY_HOOK 0\x0D
Hook VoIP On Hook Channel Name testString Party Line 5	EP testString KEY KEY_HOOK 0\x0D
Hook VoIP Redial Channel Name testString Party Line 1	EP testString KEY KEY_REDIAL\x0D
Hook VoIP Redial Channel Name testString Party Line 5	EP testString KEY KEY_REDIAL\x0D
Initiate or Add to Conference Call 0 Channel Name testString	EP testString KEY KEY_CONFERENCE 0\x0D
Initiate or Add to Conference Call 5 Channel Name testString	EP testString KEY KEY_CONFERENCE 5\x0D
Join Conference None Channel Name testString	EP testString KEY KEY_CONFERENCE\x0D
Macro None	MCCF MacroName\x0D
Matrix Tie Tie Input testString Output testString Crosspoint Type Gated	MT testString testString 1 0 3\x0D
Matrix Tie Tie Input testString Output testString Crosspoint Type Non-Gated	MT testString testString 1 0 4\x0D
Matrix Tie Tie Input testString Output testString Crosspoint Type Normal	MT testString testString 1 0 1\x0D
Matrix Tie Tie Input testString Output testString Crosspoint Type Pre-AEC	MT testString testString 1 0 5\x0D
Matrix Tie Untie Input testString Output testString Crosspoint Type Gated	MT testString testString 0 0 3\x0D

Global Scripter Module Communication Sheet

Matrix Tie Untie Input testString Output testString Crosspoint Type Non-Gated	MT testString testString 0 0 4\x0D
Matrix Tie Untie Input testString Output testString Crosspoint Type Normal	MT testString testString 0 0 1\x0D
Matrix Tie Untie Input testString Output testString Crosspoint Type Pre-AEC	MT testString testString 0 0 5\x0D
Mic Phantom Power Off Channel Name testString	EP testString LEVEL PHAN_PWR 0\x0D
Mic Phantom Power On Channel Name testString	EP testString LEVEL PHAN_PWR 1\x0D
Mute Off Channel Name testString	EP testString LEVEL MUTE 0\x0D
Mute On Channel Name testString	EP testString LEVEL MUTE 1\x0D
Output Mic/Line Line Level Channel Name testString	EP testString LEVEL MICLINE 0\x0D
Output Mic/Line Mic Level Channel Name testString	EP testString LEVEL MICLINE 1\x0D
Output Polarity Default Channel Name testString	EP testString LEVEL POLARITY 0\x0D
Output Polarity Reverse Channel Name testString	EP testString LEVEL POLARITY 1\x0D
Post Gain 0 Channel Name testString	EP testString COMPRESSOR POST_GAIN 0\x0D
Post Gain 20 Channel Name testString	EP testString COMPRESSOR POST_GAIN 20\x0D
Reject Line 1 Channel Name testString	EP testString KEY KEY_REJECT 1\x0D
Reject Line 2 Channel Name testString	EP testString KEY KEY_REJECT 2\x0D
Reject Line 3 Channel Name testString	EP testString KEY KEY_REJECT 3\x0D
Reject Line 4 Channel Name testString	EP testString KEY KEY_REJECT 4\x0D
Reject Line 5 Channel Name testString	EP testString KEY KEY_REJECT 5\x0D
Room Select None Room Number 0 Sub Room Config File testString	ROOM 0 7 testString testString\x0D
Room Select None Room Number 32767 Sub Room testString Config File testString	ROOM 32767 7 testString testString\x0D
Select Party Line 1 Channel Name testString	EP testString KEY KEY_PARTY 1\x0D
Select Party Line 5 Channel Name testString	EP testString KEY KEY_PARTY 5\x0D
Signal Generator Enable Off Channel Name testString	EP testString SIG_GEN ENABLE 0\x0D
Signal Generator Enable On Channel Name testString	EP testString SIG_GEN ENABLE 1\x0D

Appendix B. Update Commands

Beam Report Device Name testString BMA360 ID testString Beam 1	BEAM * 5 1\x0D
Beam Report Device Name testString BMA360 ID testString Beam 12	BEAM * 5 1\x0D
Beamforming Microphone Array Gain Channel Name testString	EP testString LEVEL GAIN\x0D
Beamforming Microphone Array Mode Channel Name testString	EP testString BF BF_MODE\x0D
Beamforming Microphone Array Mute Channel Name testString	EP testString LEVEL MUTE\x0D
Beamforming Microphone Array Mute LED Channel Name testString	EP testString BF BF_LED\x0D
Beamforming Microphone Array Zone Channel Name testString Zone 1	EP BFM testString BF ZONE_1\x0D
Beamforming Microphone Array Zone Channel Name testString Zone 12	EP BFM testString BF ZONE_12\x0D
Call Direction Channel Name testString	EP testString INQUIRE DIRECTION\x0D
Call Forwarding Channel Name testString	EP testString KEY KEY_FORWARD\x0D
Call State Channel Name testString Party Line 1	EP testString INQUIRE ACTIVE_PARTIES\x0D
Call State Channel Name testString Party Line 5	EP testString INQUIRE ACTIVE_PARTIES\x0D
Caller ID Channel Name testString	EP testString INQUIRE CALLER_ID\x0D
Do Not Disturb Channel Name testString	EP testString KEY KEY_DO_NOT_DISTURB\x0D
Firmware	VERSION * FW 1\x0D
Gain Channel Name testString	EP testString LEVEL GAIN\x0D
Gain Coarse Channel Name testString	EP testString LEVEL GAIN_COARSE\x0D
Gain Fine Channel Name testString	EP testString LEVEL GAIN_FINE\x0D
Gate Report Device Name testString Channel 1	GATE testString 1 1\x0D
Gate Report Device Name testString Channel 12	GATE testString 1 1\x0D
Gate Report Device Name testString Channel 4	GATE testString 1 1\x0D
Graphic EQ Enable Channel Name testString	EP testString GRAPHICEQ ENABLE\x0D
Graphic EQ Gain Channel Name testString Gain Number 1	EP testString GRAPHICEQ GAIN_1\x0D
Graphic EQ Gain Channel Name testString Gain Number 10	EP testString GRAPHICEQ GAIN_10\x0D
Hook Telco Channel Name testString	EP testString INQUIRE HOOK\x0D
Mute Channel Name testString	EP testString LEVEL MUTE\x0D
Output Mic/Line Channel Name testString	EP testString LEVEL MICLINE\x0D
Output Polarity Channel Name testString	EP testString LEVEL POLARITY\x0D
Post Gain Channel Name testString	EP testString COMPRESSOR POST_GAIN\x0D
Signal Generator Enable Channel Name testString	EP testString SIG_GEN ENABLE\x0D
Skype Call Status Channel Name testString Session ID 0	EP testString INQUIRE SESSION_CALL_STATE 0\x0D
Skype Call Status Channel Name testString Session ID 5	EP testString INQUIRE SESSION_CALL_STATE 5\x0D